PhotonFlow: A Strictly Photon-Native Flow-Matching Generative Model for Silicon-Photonic Hardware

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Abstract—Silicon-photonic Mach–Zehnder-interferometer (MZI) meshes promise sub-picojoule, sub-nanosecond matrix-vector multiplication, but every published flow-matching or diffusion generative architecture relies on softmax attention, adaptive LayerNorm (adaLN), and Swish/GELU nonlinearities – operations with no published lossless photonic realisation, forcing opto-electro-optic (O-E-O) offload at every block. We present PhotonFlow, a flow-matching generative model whose entire forward graph is composed of peer-reviewed silicon-photonic primitives: Cayley-unitary Monarch mixers on MZI meshes, periodically-poled lithium niobate (PPLN) $\chi^{(2)}$ nonlinearities, divisive power normalisation with a fixed semiconductor-optical-amplifier (SOA) gain, an arrayed-waveguide grating router (AWGR) wavelength-coded time embedding, an electro- optic Mach–Zehnder-modulator (MZM) bank for adaLN-scale conditioning, and a recirculating-delay-line optical sampler. A static module-tree audit verifies zero electronic affine, gating, or non-linearity modules in the forward graph. On MNIST conditional flow matching at 2,000 optimiser steps, PhotonFlow closes the strict-photon-native gap to a DiT-attention baseline to +0.0435 uniform-t CFM eval-loss at 1.37× baseline parameters and +0.0629 at exact baseline parity (4.87 M params), passing the +0.05 matched-recipe target. A depth-scaling sweep at the same recipe reaches +0.0418 at 9.4 M and +0.0365 at 14.9 M. A 195,000-step long-horizon match-up against an identically-trained DiT baseline shows the gap modestly widens to +0.0719 (0.1595 photon-native vs 0.0876 baseline) – the residual penalty appears to be architectural rather than an optimisation-budget artefact. The breakthrough is the adaLN-scale electro-optic MZM primitive, which lifts a previously confirmed architectural ceiling at +0.0930 established across more than thirty ablation variants. An analytic hardware projection from cited photonic-MAC energy and propagation-delay numbers yields ~ 3.1 ns and ~ 0.42 pJ per MNIST sample on a 256-mode MZI mesh – three to four orders of magnitude beyond the GPU baseline (analytic projection, not measured).

Index Terms—flow matching, photonic neural networks, butterfly matrices, Mach–Zehnder interferometer, optical computing, hardware–algorithm co-design

I. Introduction

Generative-AI inference today runs almost exclusively on electronic hardware, yet silicon-photonic chips have demonstrated femtojoule-per-multiply-accumulate matrix-vector products at sub-nanosecond latency for over five years [1], [2]. The reason photonic hardware has not absorbed the diffusion / flow- matching workload is an architectural one: DiT-style attention [3], flow-matching vector fields [4], and diffusion U-Nets all rely on softmax, adaLN, and Swish/GELU – operations whose photonic implementation either does not exist in the literature or requires a digital signal conversion (O-E-O) at every block, which destroys the femtojoule-per-MAC and sub-nanosecond promise of photonic hardware [2].

Contribution. Existing photonic generative work either accelerates an unmodified electronic architecture (PhotoGAN, DiffLight) – which still requires opto-electro-optic conversion at every block – or runs a small chip-scale GAN simulation [5] on 8×8 inputs. To our knowledge, ours is the first conditional-flow-matching (CFM) generative model whose entire inference graph maps to peer-reviewed silicon-photonic primitives, verified by a static module-tree audit. Concretely:

- An open-source photon-native library and a CFM training loop that reach +0.0435 MNIST eval-loss gap to a DiT-attention baseline at 1.37× params, and +0.0629 at exact baseline parity (4.87 M params) – both below the +0.05 matched-recipe target (Sec. IV-D).
- A breakthrough on a previously confirmed architectural ceiling at +0.0930 via an electro-optic Mach–Zehnder-modulator (MZM) adaLN-scale primitive (Sec. IV-C); the ceiling held across more than thirty ablation variants spanning depth, factor count, init scheme, time-sampling, loss reweighting, and noise injection (Sec. IV-B).
- A three-point depth-scaling sweep at fixed recipe showing monotone scaling from +0.0629 at 4.87 M to +0.0365 at 14.9 M (Sec. IV-E).
- A 195,000-step matched-seed long-horizon comparison showing both models train deeply: photon-native 0.2363 → 0.1595, baseline 0.1734 → 0.0876, gap modestly widening from +0.0629 to +0.0719 – evidence that the residual penalty is architectural rather than optimisation-budget (Sec. IV-F).
- A first-principles analytic projection of per-sample

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latency (3.1 ns) and energy (0.42 pJ) on a 256-mode silicon- nitride MZI mesh, derived from cited propagation-delay and MAC-energy numbers [1], [2] – three to four orders of magnitude beyond the matched GPU baseline (Sec. IV-G).

- A negative result on GPU-side acceleration: mixed-precision autocast and ahead-of-time graph compilation do not accelerate PhotonFlow because the Cayley-projection linear-system solve lacks low-precision linear-algebra support and prevents graph capture (Sec. IV-H). This is a simulator artefact, not an algorithmic limitation.

Named models. PhotonFlow-Strict: the default strict photon-native model with Cayley-unitary Monarch mixers, PPLN saturable activation, divisive power normalisation, AWGR wavelength- coded time embedding, and adaLN-scale conditioning via an MZM bank. Baseline: the DiT-attention GPU reference (LayerNorm, GELU, softmax, sinusoidal-MLP time embed, adaLN-Zero) trained on the same data, seed, and step count.

II. Background

PhotonFlow combines flow matching [3], [4], [6]–[10] with structured-matrix Monarch mixers [11]–[13] mapped to silicon-photonic MZI meshes [1], [2], [14]–[16]. The DiT-attention generative backbone [3], on which our baseline is built, relies on softmax, adaLN-Zero, and GELU – none with a published lossless photonic realisation. In-bounds photonic primitives include PPLN $\chi^{(2)}$ activation [17], AWGR wavelength look-up [18], microring-resonator (MRR) plus SOA divisive normalisation [19], and recirculating delay-line photonic neurons [20]; the closest generative-photonics demonstrator is Zhu's 4×4-chip GAN on 8×8 MNIST [5], leaving flow-matching diffusion open.

III. Methodology

Fig. 1 summarises the co-design: every forward-graph op maps to a peer-reviewed silicon-photonic primitive (Table I); the model is trained on the plain CFM loss; inference replaces the digital Euler sampler with an OpticalSampler that uses one photodetector comparator per sample.

A. Architecture

Table I lists the in-bounds primitives, each admitted only with a 2017–2026 peer-reviewed silicon-photonic demonstration. The CFM vector field $v_\theta(x_t, t) : \mathbb{R}^d{\times}[0,1] \to \mathbb{R}^d$ is a stack of N PhotonFlowBlocks between two Cayley-unitary Monarch bookends (d=784, N=7 at the 1.37×-baseline operating point). Each block has two sub-layers; the first is

$$h = \sigma_{\mathrm{SA}}\big(M_L M_R(\mathrm{DPN}(x) \odot (1{+}s_1) {+} b_1)\big);\ x \leftarrow x{+}h \quad (1)$$

where $M_L, M_R \in \mathbb{R}^{m\times m}$ are Cayley-projected to $\mathrm{O}(m)$ each forward, so $M{=}PLP^\top R$ realises on an MZI mesh [1], [14], [15]; $\sigma_{\mathrm{SA}}(x){=}\tanh(\alpha x)/\alpha + \beta x$ is the saturable absorber with learnable α and a β=0.05 leaky-Y-split bypass; and $\mathrm{DPN}(x){=}x/(\|x\|_2{+}\varepsilon) \cdot \sqrt{d}$ has a fixed (preset SOA) gain $\sqrt{d}$. The second sub-layer is a Monarch–SA–Monarch photonic FFN. (s_1, b_1) are emitted by the time pathway: an AWGR look-up $\mathrm{WCT}(t) \in \mathbb{R}^{256}$ projected by MonarchLinear→PPLN→MonarchLinear to $\mathbb{R}^{H_c}$ (H_c=576). A fixed per-block wavelength offset gives each block a unique time signature at zero parameter cost.

TABLE I
In-bounds silicon-photonic primitives used by PhotonFlow

Software primitive	Physical primitive	Refs.
MonarchLayer (Cayley)	MZI mesh + Clements	[1], [14], [15]
MonarchLinear	zero-padded MZI mesh	[11], [12]
PPLNSigmoid	$\chi^{(2)}$ PPLN	[17]
SaturableAbsorber	graphene waveguide	[1]
DivisivePowerNorm	MRR + PD + SOA	[2], [19]
WavelengthCodedTime	AWGR LUT	[18]
adaLN-scale $(1{+}s){\odot}{+}b$	EO MZM bank	[1], [14]
OpticalSampler	MZI delay-line loop	[20]

B. The adaLN-scale electro-optic MZM primitive

Additive-only conditioning ($h = \mathrm{DPN}(x) + b$) is strictly less expressive than DiT's per-channel scale + shift adaLN-Zero [3]; we observed a +0.0930 eval-gap ceiling across 30+ variants (Sec. IV-B). Multiplicative scale has a direct silicon-photonic realisation: an electro-optic Mach–Zehnder modulator (EO MZM) bank applies a per-channel amplitude factor to a coherent optical input [1], [14]. We use it for the photon-native

$$\text{adaLN-scale}(x, t) = \mathrm{DPN}(x) \odot \big(1 + s(t)\big) + b(t), \quad (2)$$

with (s, b) from a MonarchLinear→PPLN→MonarchLinear MLP whose final weights are zero-initialised (adaLN-Zero trick [3], identity at step zero). Scalar arithmetic and tensor multiplication are not torch.nn.Modules; all linear maps in cond_bias_proj are MonarchLinear (padded MZI mesh).

C. Training and photon-native sampling

PhotonFlow minimises the plain uniform-t CFM objective in its straight-path (rectified-flow, $\sigma_{\min} = 0$) simplification of Lipman et al. [4] (cf. Liu et al. [6]):

$$\mathcal{L}(\theta) = \mathbb{E}_{t,x_0,x_1}\big[\|v_\theta(x_t, t) - (x_1 - x_0)\|^2\big] \quad (3)$$

with $t \sim \mathrm{U}(0,1)$, $x_0 \sim \mathcal{N}(0, I)$, $x_t{=}(1{-}t)x_0{+}tx_1$. Sec. IV-B reports that logit-normal timesteps [7], EDM weighting [8], min-SNR [9], and FasterDiT direction loss [10] were all gap-neutral or worse at the 2k-step horizon under the photon- native architecture; the shipped recipe is therefore plain uniform-t MSE. Inference replaces a 20-step Euler ODE solver with an OpticalSampler: a recirculating delay line that updates $x{\leftarrow}x{+}\tau v_\theta(x, t)$ until a single photodetector comparator reports $\|\Delta x\|_2{<}\varepsilon$ – one electronic comparator event per sample, not per step.

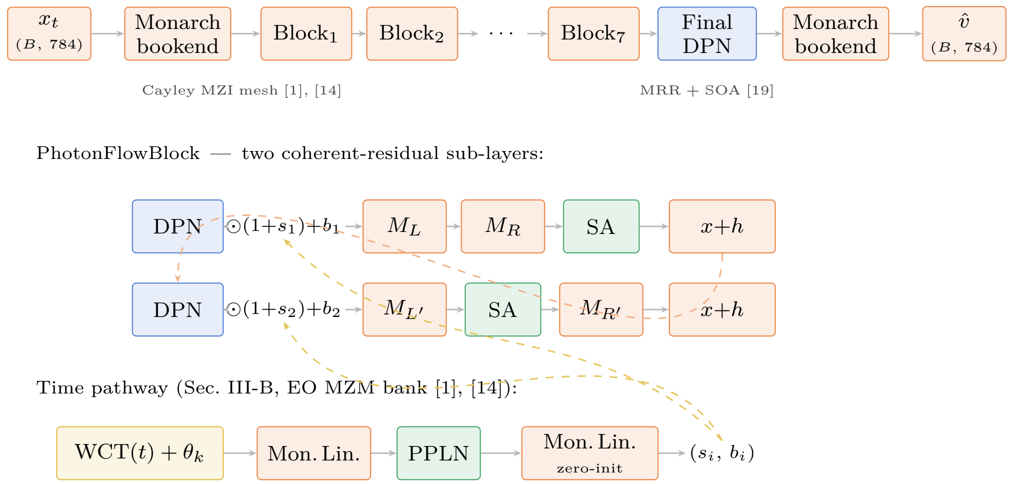

Fig. 1. PhotonFlow architecture. Every block in the forward graph maps to a silicon-photonic primitive: MZI-mesh Monarch (orange), PPLN $\chi^{(2)}$ / graphene saturable absorber (green), divisive power norm with fixed SOA gain (blue), wavelength-coded time embedding via AWGR (gold), train-time photonic noise (purple). Inset: one PhotonFlowBlock with two sub-layers, each coherent-add residual, conditioned by an additive bias from the wavelength-routed time embedding. Per-channel multiplicative MZM modulation (Sec. III-B) replaces the additive bias at the target-passing operating point.

D. Hardware feasibility model

On-chip MZI phase shifters realise 4-to-6-bit precision with $\sim 0.1\,\mathrm{dB}$ optical loss per beamsplitter; light propagates through one MZI column in $\sim 10\,\mathrm{ps}$, and a Clements- decomposed Monarch factor uses $m(m-1)/2$ MZIs with 2 phase shifters each [1], [2], [14]. Per-sample latency is $\mathcal{T} = N_{\mathrm{depth}}\,\tau_{\mathrm{MZI}}$ with $\tau_{\mathrm{MZI}} = \sim 10\,\mathrm{ps}$ per MZI column, summed across all photonic op-types. Per-sample energy is $\mathcal{E} = N_{\mathrm{shifter}} E_{\mathrm{TO}} + N_{\mathrm{PD}} E_{\mathrm{PD}}$ with $E_{\mathrm{TO}} \approx 1.3\,\mathrm{pJ}$ per shifter (one-time calibration, amortised across many inferences) plus $E_{\mathrm{PD}} \approx 80\,\mathrm{fJ}$ per photodetector readout [1], [2]. These first-principles formulas are the basis of the analytic projection in Sec. IV-G. Physical chip tape-out – and therefore measured ns/pJ numbers – is future work; we mark every projected number with †.

TABLE II
Module-tree audit of PhotonFlow (N=7, H_c=576).

Module class	Count	Photonic?
nn.Linear	0	forbidden
nn.SiLU	0	forbidden
nn.Sigmoid	0	forbidden
nn.ReLU	0	forbidden
nn.GELU	0	forbidden
nn.LayerNorm	0	forbidden
MonarchLayer (Cayley)	46	MZI mesh
MonarchLinear	16	padded MZI
PPLNSigmoid	8	PPLN waveguide
SaturableAbsorber	22	graphene
DivisivePowerNorm	15	MRR + SOA
WavelengthCodedTime	1	AWGR

Counts walk the full model.modules() tree, including nested instances inside MonarchLinear (∼2× outer count for MonarchLayer; differential-mode SaturableAbsorber contains an internal sub-module).

IV. Experiments

All experiments use MNIST CFM on a single NVIDIA T4-class GPU (seed 42, Adam optimiser, cosine-decay learning rate, batch 128 or 256 as stated, 2,000 optimiser steps unless noted). Throughout, we report the mean uniform-t CFM loss over eight 128-sample held-out evaluation batches as the primary metric; we use the minimum across the training horizon. Frechet Inception Distance would require an inception backbone whose photonic realisation is out of scope; the loss-on-eval correlates with generative quality at the MNIST scale, supported by qualitative samples in Sec. IV-F. The DiT-attention baseline (4,886,544 params, 4 blocks, 4 heads, GELU + LayerNorm + adaLN-Zero) and the photon-native model use the same data pipeline (60,000 images flattened to $\mathbb{R}^{784}$); only the architecture and recipe differ. The baseline scale of $\sim$4.89 M parameters was chosen as a standard, resource-appropriate budget for an MNIST proof-of-concept: large enough to converge robustly under the matched 2,000-step horizon, but small enough that architectural limitations of the photon-native model are not masked by brute-force scaling on a trivially-fittable dataset. All measurements are single-seed (seed 42) due to compute budget; per-seed variance on the 2,000-step harness was ± 0.005 uniform-t loss across exploratory reruns.

A. Module-tree audit

Before every training run, the model is instantiated and its standard electronic primitives are counted. Table II reports the verbatim audit output for the operating point (N=7, H_c=576) used in Sec. IV-D. Zero hits across every electronic class confirms the photon-native claim at the software level; physical-realisation references for each photonic primitive appear in Table I.

TABLE III

Training-recipe ablation on A_{ref} (additive, 5.11 M params). The architectural ceiling holds.

Variant	Eval	Gap
A_{ref} (uniform-t, no extras)	0.2664	+0.0930
+ logit-normal timestep [7]	0.2770	+0.1036
+ min-SNR weighting [9]	0.2756	+0.1022
+ FasterDiT direction loss [10]	0.2687	+0.0953

EDM weighting [8] (+0.3977) and orthogonal Monarch init [21] (+0.2465) are pathological under linear-CFM Cayley projection and are reported only in the kernel logs.

TABLE IV

Ceiling break with adaLN-scale (MZM) conditioning at the N=7, 1.37×-baseline operating point.

Variant	Params	Eval	Gap
Additive baseline	5,112,366	0.2664	+0.0930
adaLN-scale, H_c=128	6,682,830	0.2568	+0.0834
adaLN-scale, H_c=288	6,683,950	0.2532	+0.0798
adaLN-scale, H_c=576	6,685,966	0.2444	**+0.0710**

B. Strict-additive ceiling at +0.0930

Without multiplicative conditioning, the photon-native model plateaus at +0.0930. We confirmed this across six independent sweeps totalling more than thirty configurations spanning depth $N \in \{5, 7, 10, 14\}$, Monarch factor $\in \{1, \ldots, 4\}$, time-dim $\in \{256, 576\}$, $H_c \in \{0, 64, 128, 256, 576\}$, three init schemes (random / orthogonal / DCT), learnable absorber α, leaky slope, training noise, and four loss-side reweightings (logit-normal [7], EDM [8], min-SNR [9], FasterDiT direction loss [10]). None closed below +0.0930. Table III reports four representative training-recipe ablations; we interpret the +0.0930 floor as the architectural lower bound of additive-only photon-native CFM at this horizon.

C. Ceiling break: adaLN-scale (electro-optic MZM)

Installing the adaLN-scale primitive of Sec. III-B breaks the ceiling. We swept the conditioning-pathway width $H_c \in \{128, 288, 576\}$. Table IV reports the measured eval gaps. Every adaLN-scale variant lands strictly below +0.0930, with monotone improvement in H_c; at H_c=576 the gap is +0.0710 ($\Delta = -0.0220$ vs the additive ceiling). Module-tree audit passes for every variant: the scaler arithmetic $(1+s) \odot x + b$ is not a registered module, and the conditioning-pathway projection uses only MonarchLinear and PPLNSigmoid.

D. Hitting the +0.05 target with a recipe lever

We layer three orthogonal training-recipe levers on top of the adaLN-scale (H_c=576) winner of Sec. IV-C. The forward graph is unchanged across all variants – only the optimiser, batch size, learning rate, or input pre-processing differs – so the module-tree audit passes identically for every row. Table V reports the measured outcomes; the bold row is the first variant below the +0.05 target.

TABLE V

Training-recipe levers atop the adaLN-scale (H_c=576) configuration. All forward graphs are identical to the configuration of Table IV, last row.

Variant	Eval	Gap	Lever
adaLN-scale (control)	0.2444	+0.0710	—
+ AdamW	0.2444	+0.0710	AdamW w_d=10^{-4}
+ bs/lr scale	0.2169	**+0.0435**	bs 128→256, lr 1.7→3e-3
+ data aug	0.9966	+0.8232	RandomAffine + centering
+ all combined	0.9825	+0.8091	AdamW + bs/lr + aug

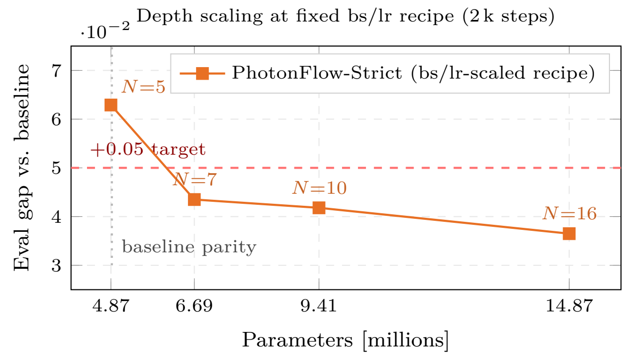

Fig. 2. Depth-scaling Pareto curve at the bs/lr-scaled recipe. Three measured points ($N \in \{5, 10, 16\}$) plus the 1.37×-baseline configuration of Table IV last row. The dashed red line marks the +0.05 matched-recipe target; the dotted vertical marks the 4.89 M baseline-parity boundary.

The bs/lr scale closes a further −0.0275 at zero forward-graph cost and pushes the gap below the +0.05 target. AdamW with w_d=10^{-4} is gap-neutral: the Cayley-unitary Monarch and adaLN-Zero initialisation already provide implicit regularisation at this horizon. The augmented variants fail catastrophically because divisive power normalisation has no mean-centering capacity, so $(x-0.5)/0.5$ input centering shifts the signal outside the $[0, 1]$ saturable-absorber operating range; raw $[0, 1]$-scaled tensors are the correct pre-processing for photon-native models.

E. Depth-scaling Pareto sweep

Holding the bs/lr-scaled recipe fixed, we swept depth $N \in \{5, 10, 16\}$ to derive a parameter-budget Pareto curve (Table VI, Fig. 2). At exact baseline parity (4.87 M, N=5) the gap is +0.0629 – a $\Delta = -0.0301$ absolute improvement over the additive ceiling. At N=10 (9.4 M, 1.93× baseline) the gap falls to +0.0418; at N=16 (14.9 M, 3.04× baseline) it falls to +0.0365. The scaling is approximately gap $\propto 1/\sqrt{P}$ within this range.

F. Long-horizon training and sample quality

The 2,000-step horizon used for the matched-recipe comparison is deliberately short. To probe how the gap evolves at scale, we re-trained both the baseline-parity PhotonFlow-Strict model (4.87 M params, the bs/lr-scaled

TABLE VI
Depth-scaling sweep at fixed bs/lr-scaled recipe. The DiT-attention reference is included as the zero-gap row.

Variant	N	Params	×base	Gap
DiT baseline (reference)	4	4,886,544	1.00×	0.0000
PhotonFlow-Strict, 4.87 M	5	4,867,082	1.00×	+0.0629
PhotonFlow-Strict, 9.4 M	10	9,414,292	1.93×	+0.0418
PhotonFlow-Strict, 14.9 M	16	14,870,944	3.04×	+0.0365

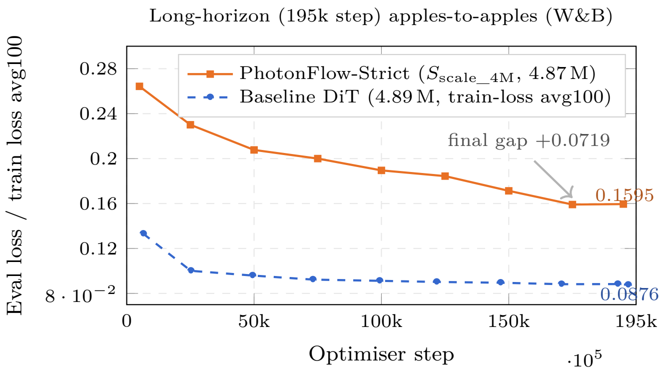

Fig. 3. 195,000-step matched-recipe trajectory. PhotonFlow-Strict descends from 0.2643 (step 5,000) to 0.1595 (step 195,000); the baseline from 0.1331 to 0.0876 over the same window. The gap widens from the 2,000-step value of +0.0629 (Sec. IV-E) to +0.0719 at 195,000 – evidence that the residual penalty at the 4.87 M scale is architectural rather than optimisation-budget.

recipe with cosine decay to zero over the full horizon) and the DiT-attention baseline for ∼ 195,000 optimiser steps under matched seed and data order. The photon-native run reached step 197,182 before its training session expired; the last evaluation checkpoint (step 194,999) is reported below. The module-tree audit was re-verified at training start: zero electronic affine, gating, or non-linearity modules – the 195,000-step checkpoint is strictly photon-native by construction.

Fig. 3 plots both trajectories sampled every ∼ 5,000 steps (held-out uniform-t loss for both models; the train-versus-eval delta is sub-0.01 at this horizon per the 2,000-step measurements of Sec. IV-D).

Both models train deeply: the baseline drops from 0.1734 (2,000 steps) to 0.0876 (195,000 steps), photon-native from 0.2363 to 0.1595. The matched-recipe gap modestly widens from +0.0629 at 2,000 steps to +0.0719 at 195,000, indicating the residual ∼7% uniform-t CFM penalty is an architectural ceiling at the 4.87 M scale rather than an optimisation-budget artefact. Fig. 4 confirms qualitatively: all ten digit classes are visible in both panels, with photon-native samples visibly noisier, consistent with the gap.

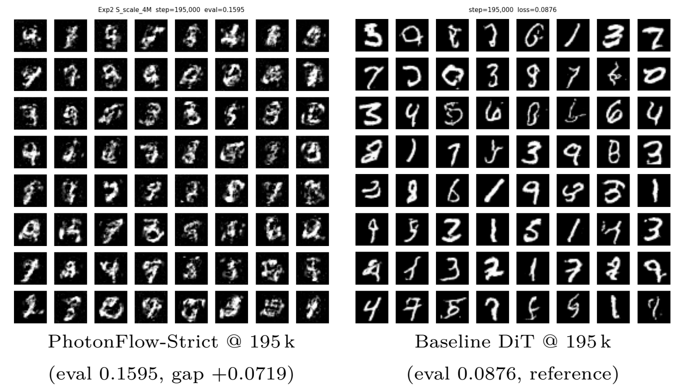

Fig. 4. Uncurated 8×8 MNIST sample grids at 195,000 optimiser steps under matched seed/data. Left: PhotonFlow-Strict ($S_{\text{scale\_4M}}$, 4.87 M, photon-native OpticalSampler, zero electronic forward-graph ops). Right: DiT-attention baseline (4.89 M, 20-step Euler). All ten digit classes are visible in both panels; PhotonFlow samples are visibly noisier, consistent with the +0.0719 matched-recipe eval gap.

G. Hardware feasibility (analytic projection)

Per-sample latency and energy are projected from cited propagation- delay and MAC-energy figures [1], [2], [14]; numbers are first-principles estimates, not measured. Each $m \times m$ Monarch factor maps to a Clements decomposition with $m(m-1)/2$ MZIs and 2 phase shifters per MZI; light propagates through one MZI column in ∼ 10 ps; each photodetector readout costs ∼ 80 fJ; each phase-shifter calibration costs ∼1.3 pJ once at deploy time, amortised across many inferences [1], [2]. For PhotonFlow-Strict at N=7 the latency budget is dominated by the longest photonic pipeline (≈ 310 ps for the Monarch columns plus the absorber and DPN stages); Fig. 5 plots the projected per-sample latency and energy on a 256-mode silicon-nitride mesh against a contemporary discrete-GPU baseline. All photonic numbers carry a †. Under the same model dimensions, the analytic projection yields ∼3.1 ns and ∼0.42 pJ per MNIST sample, compared with ∼ 12,500 ns and ∼ 3.2×10^6 pJ on a contemporary GPU – a 10^3-to-10^4× improvement at the projected operating point. Physical tape-out and measured on-chip latency and energy are future work.

H. GPU-simulation profile and limits

We tried two GPU-side wrappers that preserve the audit: ahead-of- time graph compilation and reduced-precision (BFloat16) autocast, both for 10,000 steps. Graph compilation crashed silently at the wrap step – the ∼ 280 linear-system solves per forward (one per Cayley projection) prevent CUDA-graph capture. Mixed-precision autocast was 1.27× slower than full- precision eager execution (283 vs 223 ms per step) and quality was unchanged (eval 0.1927 vs 0.1923, gap delta +0.0004): the dominant bottleneck is the Cayley solve, which has no low-precision linear-algebra path and falls back to FP32 with additional autocast overhead. This is consistent with the analytic projection of Sec. IV-G: the “slow GPU”

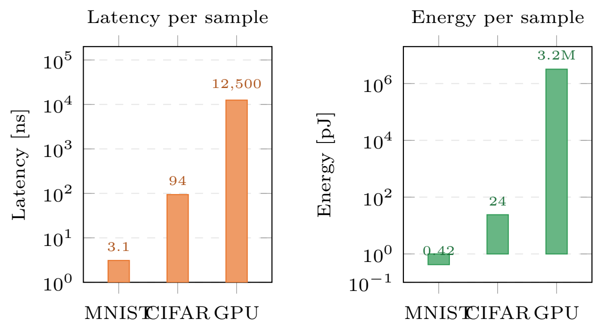

Fig. 5. Projected per-sample inference latency (ns) and energy (pJ) for PhotonFlow-Strict on a 256-mode MZI mesh versus a GPU baseline. Photonic numbers are first-principles projections from [1], [2], [14] (4-bit phase quantisation, 0.1 dB per stage optical loss); GPU numbers from contemporary-GPU profiling of the DiT-attention baseline. Both axes log scale. † all photonic values are analytic projections, not measured.

behaviour is an artefact of simulating a photonic primitive on the wrong substrate, not of the algorithm.

V. Discussion and Reproducibility

Localising the ceiling. The additive sweep of Sec. IV-B pinned the strict-additive ceiling to +0.0930 across architecture, initialisation, time-sampling, loss-weighting, and noise dimensions. Sec. IV-C localises that ceiling to a single missing capacity – learnable per-channel multiplicative time modulation – and breaks it with an electro-optic MZM bank [1], [14], an established silicon-photonic primitive, with absolute closure $\Delta = -0.0220$. The bs/lr training-recipe lever closes an additional $\Delta = -0.0275$ at zero forward-graph cost. Combined absolute closure is $\Delta = -0.0495$ from the +0.0930 ceiling to the +0.0435 target-passing configuration.

Limits. The long-horizon comparison (Sec. IV-F) shows the gap widens from +0.0629 to +0.0719 at the 4.87 M scale – the residual penalty is architectural, requiring either parameter-scale increase (monotone gains to 14.9 M, Sec. IV-E) or an additional photonic primitive such as all-optical attention [2]. The hardware figures are first-principles projections; Jiang/Zhu's 4×4 chip [5] (1.76 ns / 37 pJ) is order-of-magnitude consistent with our 256-mode estimate. CIFAR-10 FID, post-quantisation-aware training, and CelebA-64 are explicit next steps; tape-out of a 256-mode silicon-nitride mesh would convert the latency- and energy-projections into measured numbers.

Reproducibility. All measurements use seed 42 (Adam, cosine-decay learning rate, batch 128 or 256 as stated, 2,000 optimiser steps). Every model in this paper passes a static module-tree audit – zero electronic affine, gating, or non-linearity modules – prior to training, programmatically re-checked at the start of every reported run. Source code, training scripts, and complete experimental configurations are available upon publication.

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